

Career Aspirations And Preparedness



Introduction

Career aspirations and preparedness among students are critical factors shaping their future professional trajectories. Understanding the influences, motivations, and challenges faced by undergraduates can provide valuable insights into the support systems and guidance they require. This particular survey was done for the 3rd year undergraduates of the Faculty of Science, University of Colombo.



Objectives

- ❑ To understand students' career goals, the factors influencing them, their level of readiness, and the challenges they face as they prepare for their career goals
- ❑ Investigate how career goals are influenced by degree type
- ❑ Analyze the relationship between career preparedness and the frequency of knowledge-seeking activities.
- ❑ Evaluate the relevance of academic skills to career aspirations across different degree types
- ❑ Identify the challenges faced by students in achieving their career goals and examine how factors such as gender, hometown area, and degree type related with these challenges and sector preferences.



Data

	Gender	Hometown_Area	Degree_Type	GPA	GPA_Credits	Career_Goal	Sector_Preference	Career_Influences	Additional_Knowledge	Knowledge_Sources	Knowledge_Frequency	Skill_Relevance	Career_Preparedness	Career_Experiences	Guidance_Usage	Guidance_Satisfaction	Career_C
0	Female	Semi Urban	Industry Oriented Honors degree	3.55	84	Secure employment within the home country	Privat sector	Personal interests/passion;Job market trends a...	Yes	Do online courses;Explore online articles;	Monthly	Very relevant	Somewhat prepared	None;	Yes	Very satisfied	Insufficie qualifications;C
1	Male	Semi Urban	Industry Oriented Honors degree	3.60	66	Secure employment within the home country	Privat sector	Personal interests/passion;Job market trends a...	Yes	Do online courses;	Monthly	Very relevant	Somewhat prepared	Internships;Volunteering or community service;	Yes	Very satisfied	L opportuni desi
2	Female	Semi Urban	Industry Oriented Honors degree	3.40	75	Seek employment opportunities abroad	NaN	Personal interests/passion;Potential salary an...	Yes	Explore online articles	Monthly	Very relevant	Neither prepared nor unprepared	NaN	No	NaN	Lack of gu r
3	Female	Urban	Industry Oriented Honors degree	3.68	84	Secure employment within the home country	Privat sector	Personal interests/passion;Job market trends a...	Yes	Do online courses	Monthly	Somewhat relevant	Somewhat prepared	NaN	Yes	Somewhat satisfied	Insufficie qua
4	Male	Semi Urban	Research Oriented Honors degree	3.90	75	Pursue higher education (e.g., master's, PhD)	NaN	Personal interests/passion;Job market trends a...	No	NaN	NaN	Very relevant	Very prepared	Part-time or freelance work in a related field	Yes	Somewhat satisfied	Financial c

- This dataset contains **17** features. Every column corresponds to a question in the questionnaire.
- A total of **152 responses** were collected from the target population.
- A post-stratification sampling method was used to adjust the sample proportions according to the population distribution, based on the variable ‘Degree Type’, a sample of **120 responses** was selected.

Data

Dataset Features

Categorical Features

- Gender, Hometown Area, Degree Type, Career Goal, Sector Preference, Career Influences, Additional Knowledge, Knowledge Sources, Knowledge Frequency, Skill Relevance, Career Preparedness, Career Experiences, Guidance Usage, Guidance Satisfaction, Career Challenges

Numerical Features

- GPA, GPA Credits

Original Dataset:-

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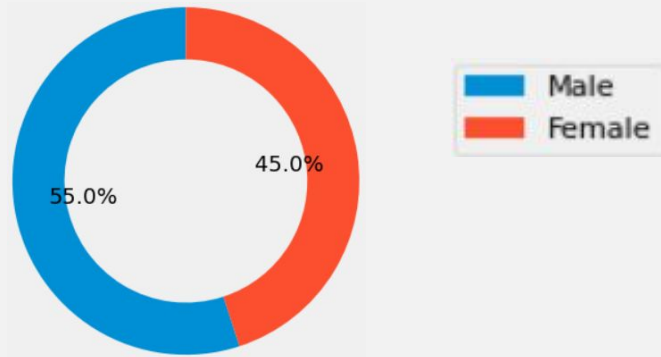
Sampled Dataset:-

https://drive.google.com/file/d/1NeEmqOFbtRfytpMP-fVOy2AEVVn9_YJ0/view?usp=drive_link

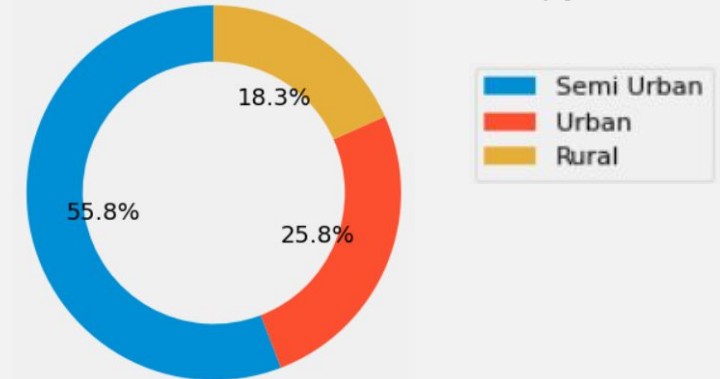
Descriptive Analysis (Univariate)

Demographic Information

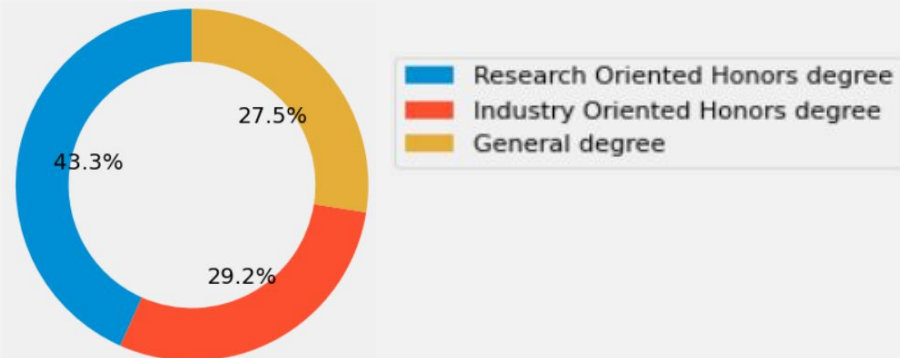
Gender



Hometown Area Type

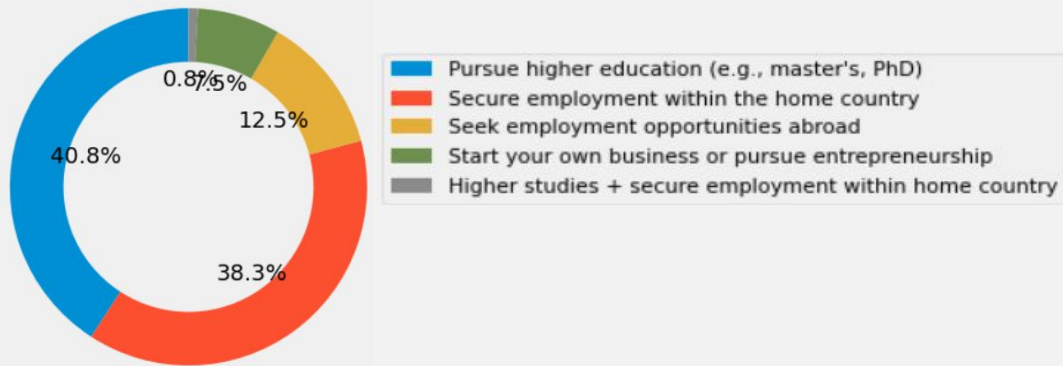


Degree Type

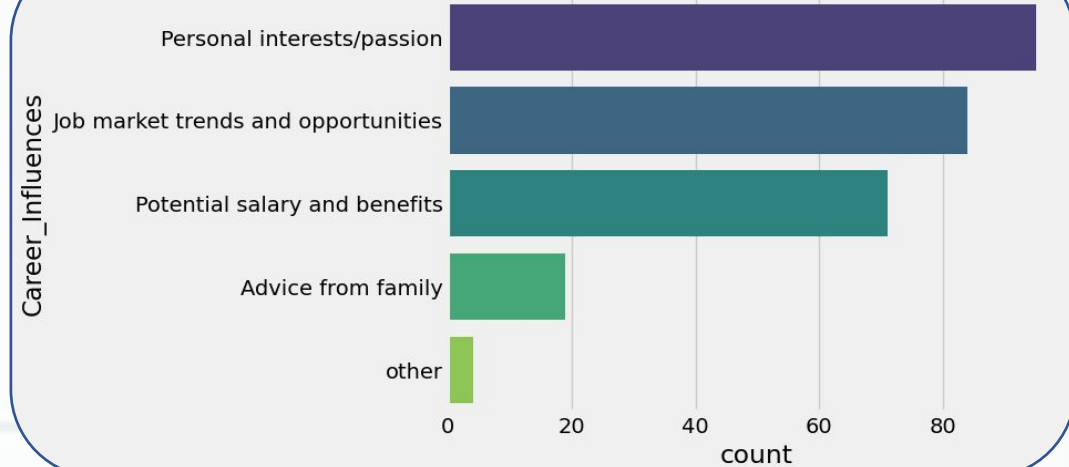


Descriptive Analysis (Univariate)

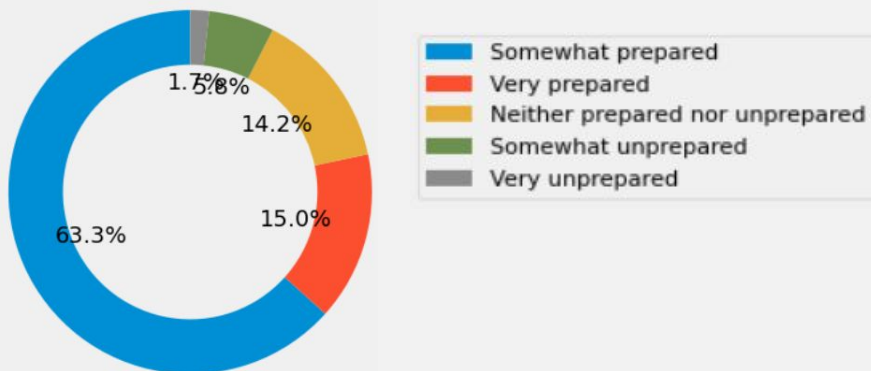
Career Goals



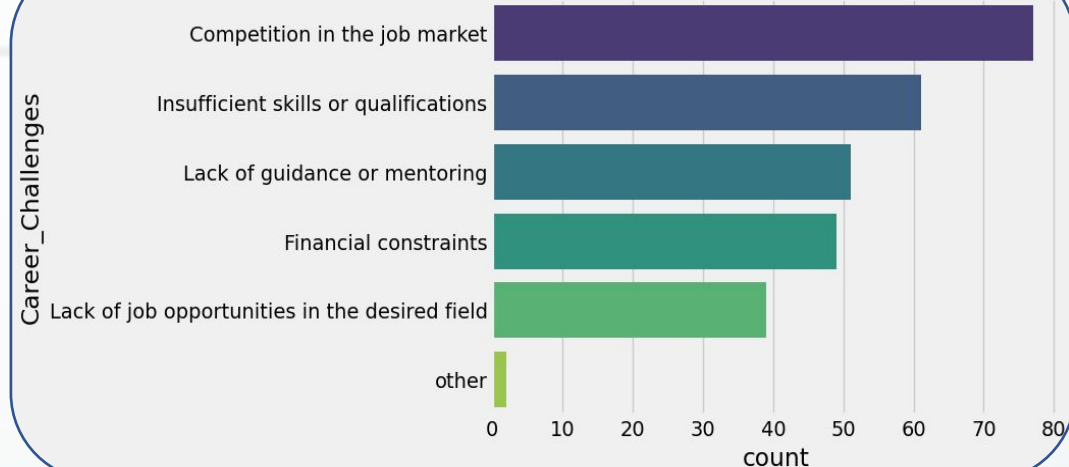
Career Influences



Career Preparedness

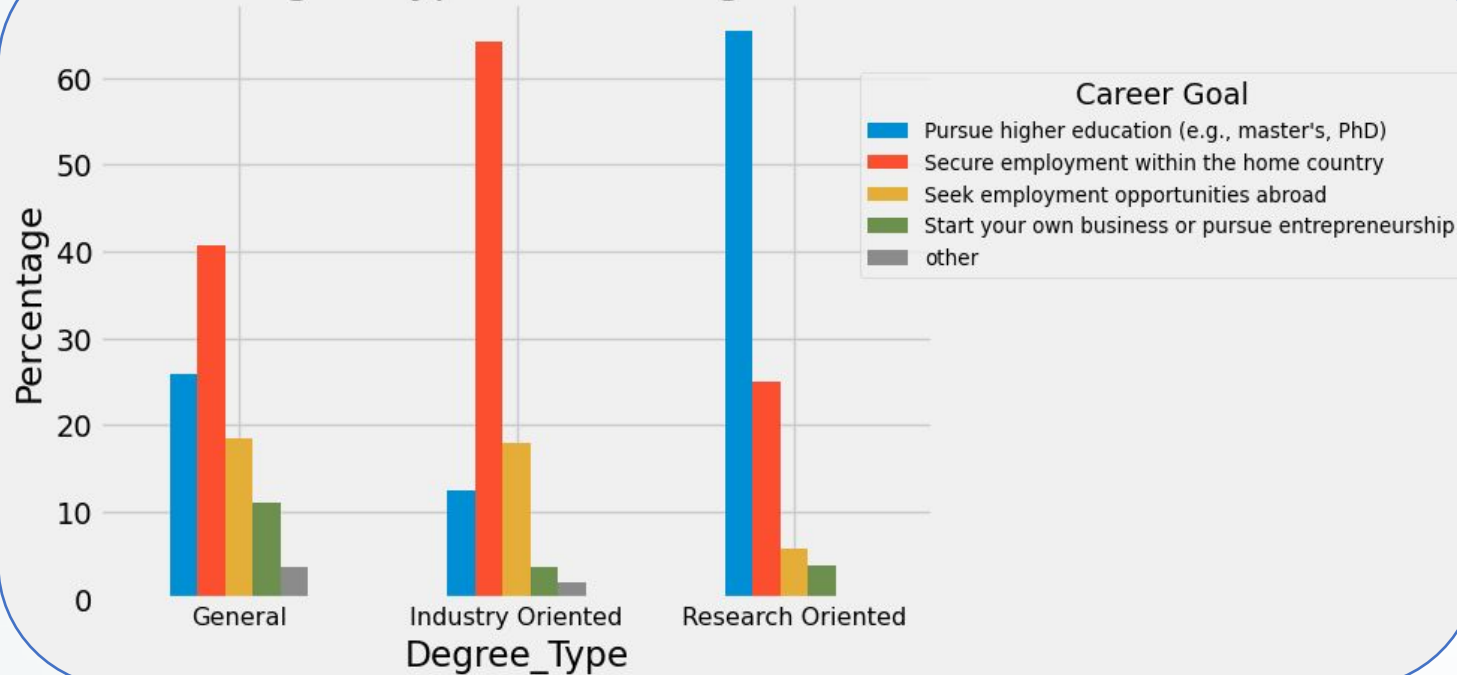


Career Challenges



Descriptive Analysis (Bivariate)

Degree type Vs Career goal



Out of the students who follow research oriented degree programme, the majority of them plan to pursue higher education

Majority of the students from the other two degree programmes want to secure employment within the home country

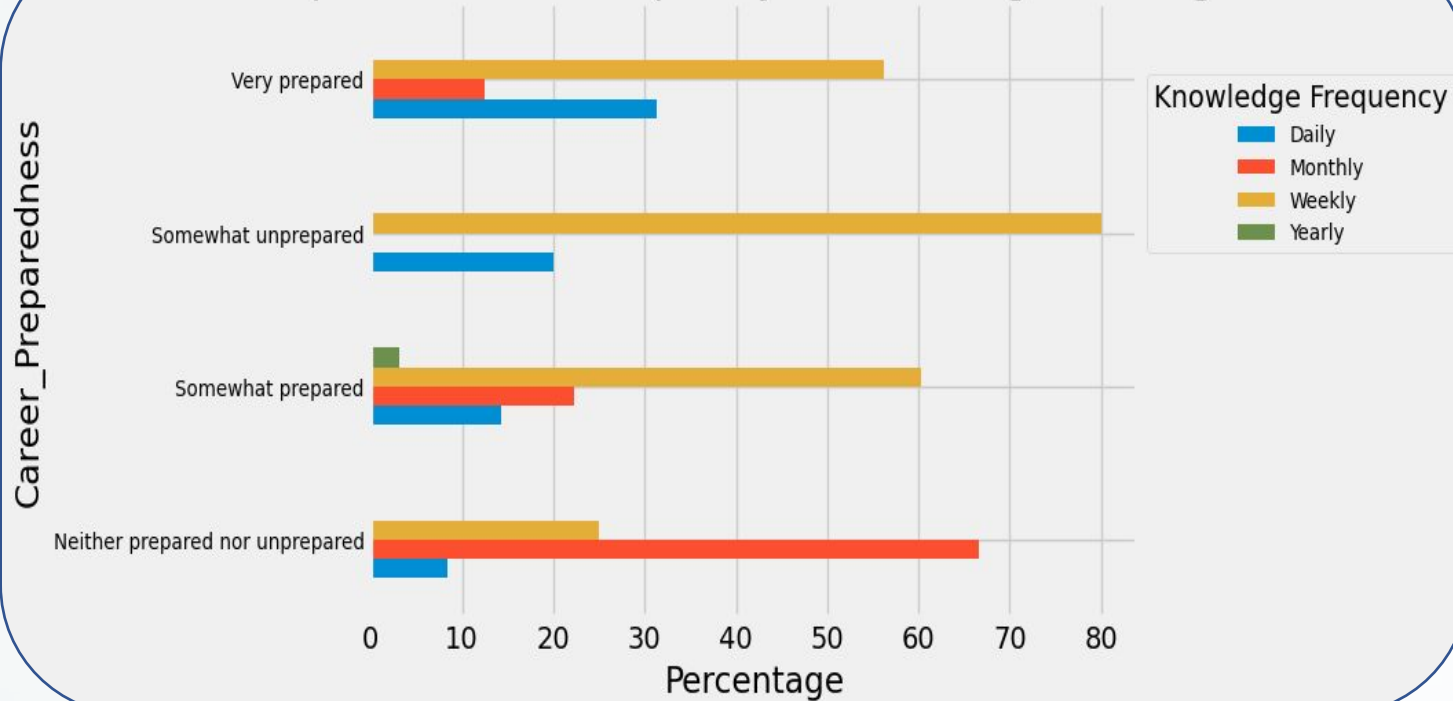
Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	29.559 ^a	4	.000
Likelihood Ratio	30.531	4	.000
N of Valid Cases	120		

Since the p-value is less than 0.05, we can confirm that there is a relationship between degree type and career goal

Descriptive Analysis (Bivariate)

Career Preparedness Vs Frequency of Knowledge seeking activities



Students who mentioned 'very prepared' for their career, engage in knowledge seeking activities more often.

Majority of students who mentioned 'Neither prepared or unprepared', engage in knowledge seeking activities monthly

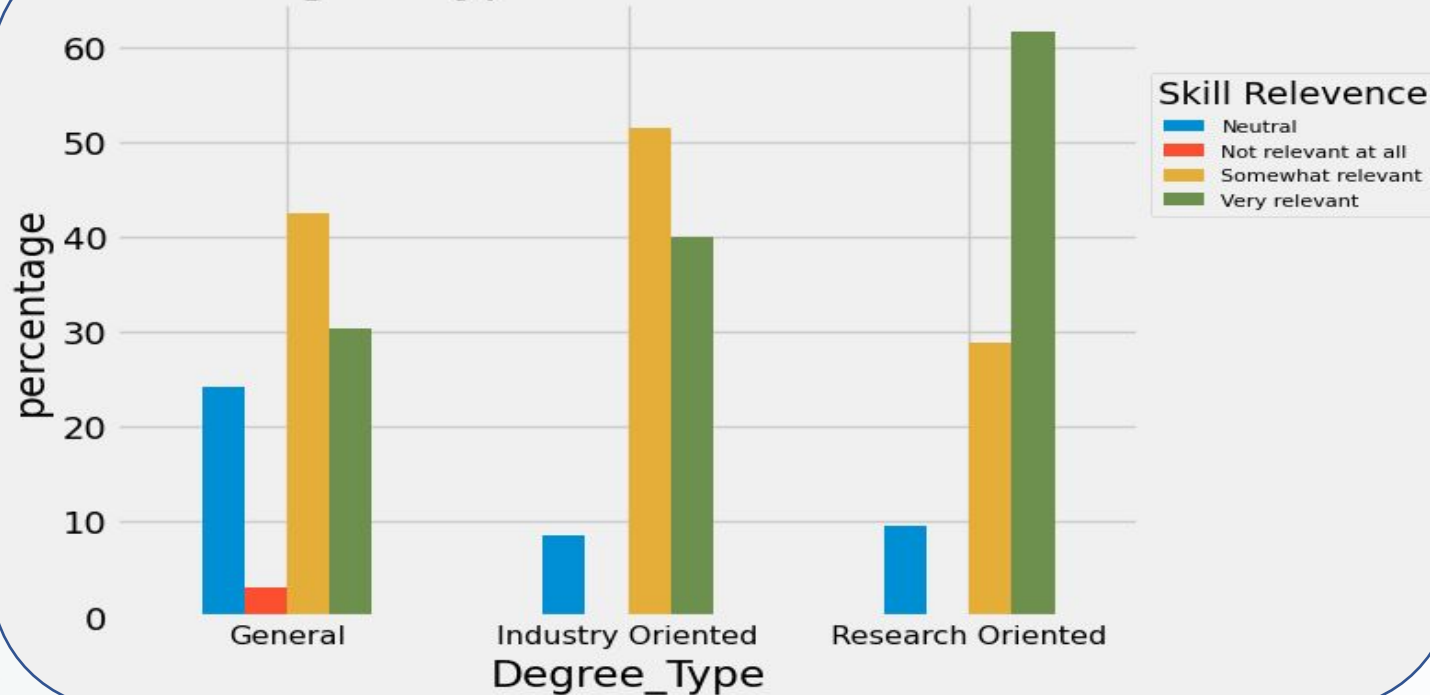
Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	4.174 ^a	1	.041		
Continuity Correction ^b	3.035	1	.081		
Likelihood Ratio	3.844	1	.050		
Fisher's Exact Test				.068	.044
N of Valid Cases	96				

But as the p-value is greater than 0.05, we do not reject the null hypothesis that there's no significant association between the career preparedness and frequency of knowledge seeking

Descriptive Analysis (Bivariate)

Degree Type Vs Skill Relevance



Majority of students who are doing research-oriented degree programmes, believe their skills are very relevant to their career goals.

While majority of students from other two degree programmes believe their skills are somewhat relevant, the neutral percentage is high in general degree

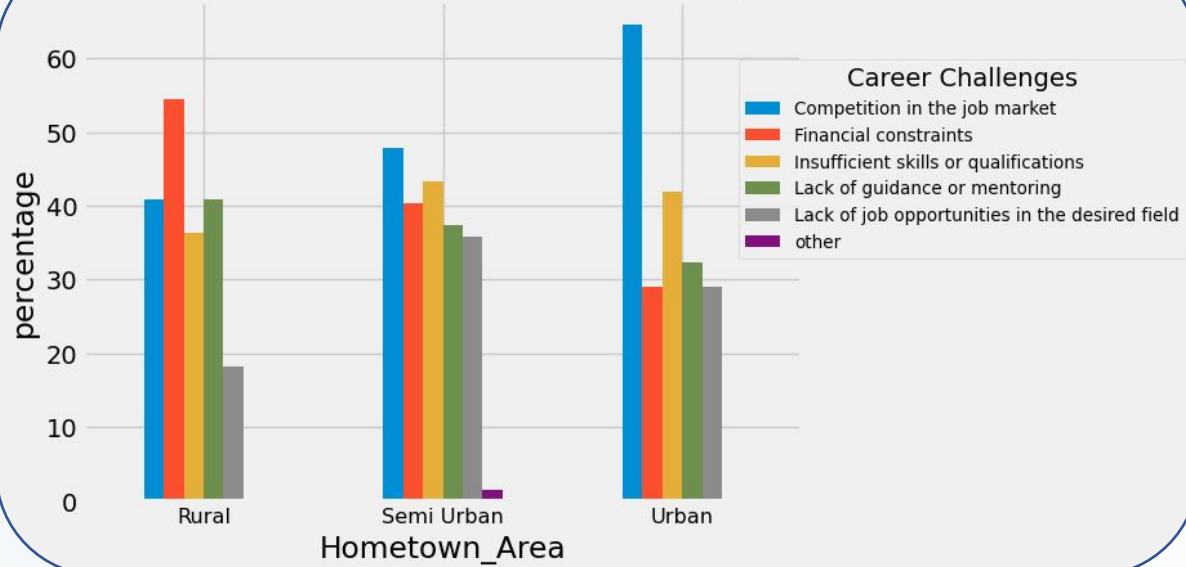
Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	6.449 ^a	2	.040
Likelihood Ratio	5.845	2	.054
N of Valid Cases	120		

Since the p-value is less than 0.05, we can confirm that there is a relationship between degree type and skill relevance

Descriptive Analysis (Bivariate)

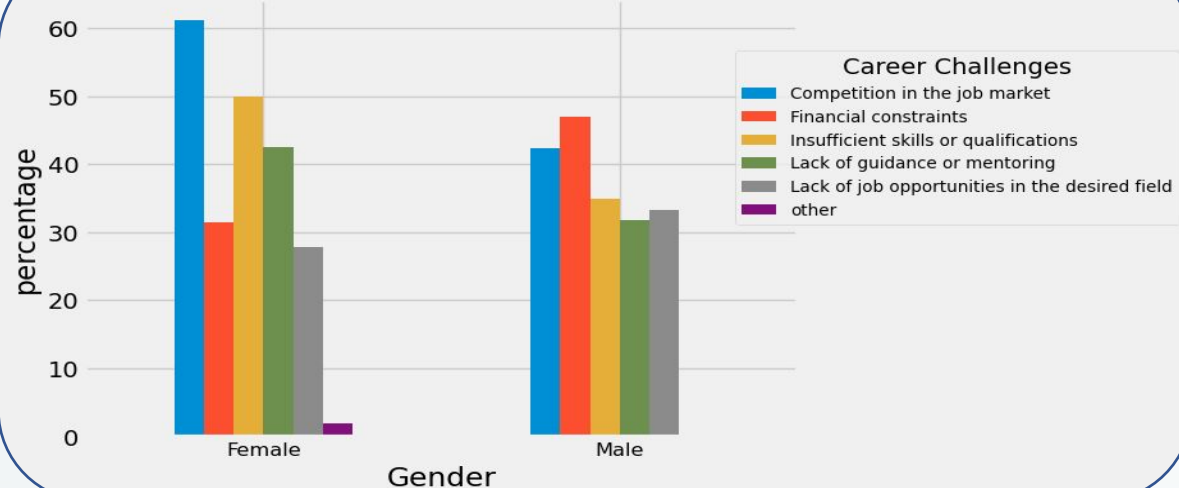
Hometown Area Vs Career Challenges



The majority of students from urban areas stated that their primary challenge is competition in the job market. So as students from semi-urban areas

The primary challenge of rural area students is financial constraints.

Gender Vs Career Challenges



Female students identify competition in the job market as their primary challenge, while male students mentioned financial constraints as their main obstacle.

Conclusion

- ✓ Students pursuing research-oriented degree programs primarily aim to pursue higher education, while others prefer securing local employment.
- ✓ We have evidence that frequent knowledge-seeking activities may enhance career preparedness, although the relationship was not significant
- ✓ Research-oriented students find their skills highly relevant, while others believe them as “somewhat relevant.” Programs emphasizing practical skills and career alignment are especially valuable for students in the other two degree programmes
- ✓ Urban and semi-urban students need strategies to navigate competitive job markets, while rural students may benefit from financial aid and scholarships.
- ✓ Female students primarily face challenges related to competition in the job market, while male students report financial constraints as their major obstacle.



Limitations

- ❑ Only 120 responses were analysed, which may not fully represent the entire population.
- ❑ Non-response bias may have affected the results as certain students chose not to participate.
- ❑ Career aspirations and preparedness are subjective topics, and some variables may not fully capture the complexity of students' experiences or future plans.
- ❑ The limited timeframe for conducting the survey may have restricted the amount of detailed information that could be gathered



Limitation

Future Work

- ❑ Conduct surveys across multiple faculties or universities in Sri Lanka to gain a broader perspective on career aspirations.
- ❑ Track the same group of students over time, from graduation to post-graduation, to analyze changes in aspirations and preparedness.
- ❑ Use the findings to propose improvements to the career guidance programs offered by the faculty.
- ❑ Incorporate focus group discussions or interviews to gain deeper insights into career influence factors and preparedness.



THANK
YOU

